Answer **ALL** questions

1 When a glycoprotein is being synthesized for secretion from a cell, which route is it most likely to take?

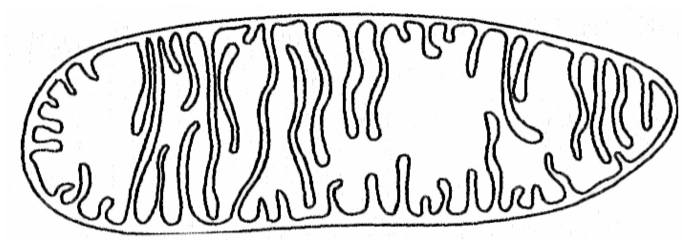
A golgi apparatus → rough endoplasmic reticulum → smooth endoplasmic reticulum

B rough endoplasmic reticulum → golgi apparatus → smooth endoplasmic reticulum

C smooth endoplasmic reticulum → rough endoplasmic reticulum → golgi apparatus

D rough endoplasmic reticulum → smooth endoplasmic reticulum → golgi apparatus

2 The diagram shows an organelle found in a plant cell.



Which molecules are found in this organelle?

A cellulose and protein

B phospholipid and protein

C triglyceride and glucose

D starch and glucose

3 Which one of the following statements is **INCORRECT**?

A There are 3 polypeptide chains in the collagen helix, and only one in the α-helix.

B Proline is a major component of collagen but not of the α-helix.

C Every third residue in the α-helix must be glycine, but there is no sequence regularity in collagen.

D Some hydrogen bonding exists between different chains in collagen, but hydrogen bonding occurs within the same chain in the α-helix.

4 In the diagram below, the numbers **1** – **3** refer to different amino acids and the letters **X** and **Y** refer to different enzymes.

$$\mathbf{1} \xrightarrow{\mathbf{X}} \mathbf{2} \xrightarrow{\mathbf{Y}} \mathbf{3}$$

The artificial introduction into the cell of an excess of amino acid 3 reduces the rate of reaction catalysed by enzyme Y. What is the cause of this?

A Competitive inhibition

B End-product inhibition

C Positive feedback

D Enzyme denaturation

5 Succinic dehydrogenase is the enzyme that catalyses the oxidation of succinic acid during cellular respiration. If malonic acid is added to the system, the rate of reaction is reduced.

An increase in substrate concentration will, however, increase the rate of reaction again. From the above information, what can you conclude about malonic acid?

A It binds to the allosteric site of the enzyme.

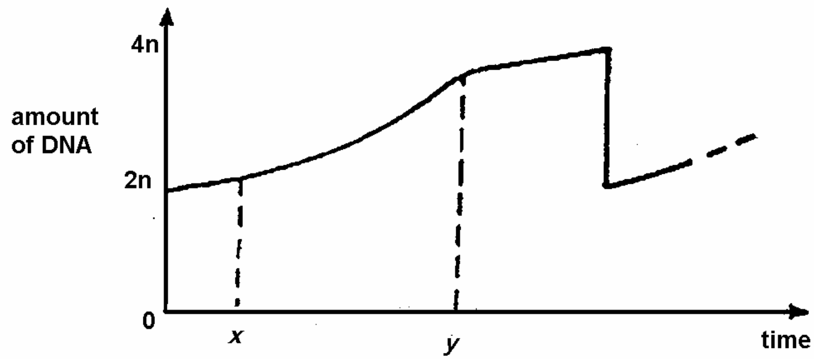
B It increases the pH of the system.

C It has a similar molecular configuration to that of succinic acid.

D It forms a permanent attachment to the active site of the enzyme.

6 The graph below shows the changes in DNA content in a single cell during the mitotic cycle.

Which stage occurs between time **x** and time **y**?



A telophase

B interphase

C metaphase

D anaphase

7 Throughout which phase of the mitotic cycle of a single cell or daughter cell is DNA present in the least amount?

A Anaphase

B Interphase

C Prophase

D Telophase

8 Which of the following enzymes or proteins is **NOT** used in the process of DNA replication?

A Primase

B DNA polymerase

C Translocase

D Single-strand binding protein

9 How many different polypeptides, each consisting of 3 amino acids, can be made if the number of different amino acids available is 10?

A 30

B 1000

C 3000

D 59049

10 Below is part of the DNA genetic code for six amino acids.

CGG codes for alanine TTT codes for lysine
GCG codes for arginine AAA codes for phenylalanine
CCA codes for glycine CAA codes for valine

The diagram shows part of a mRNA molecule, but the corresponding part of the protein formed was found to be:

arginine–glycine–lysine–valine–alanine

C G C G U U A A A G U U G C C

Which triplet contains a single transcription error?

A the first

B the second

C the third

D the fourth

11 Which of the following statements is **INCORRECT**?

A Initiation of transcription requires binding of RNA polymerase to a promoter.

B All primary transcripts (RNA) are processed before being used for translation into protein.

C Double-stranded DNA, not single-stranded DNA, is used for transcription by RNA polymerase.

D Uracil, not thymine, is incorporated into RNA during transcription.

12 A gorilla mRNA differs from *E. coli* mRNA in that

A the 5' end of the gorilla mRNA is processed.

B the 3' end of the *E. coli* mRNA is processed.

C only the gorilla mRNA contains a start codon.

D only the gorilla mRNA can be translated into a polypeptide.

13 Which statement is **not** true about proto-oncogenes?

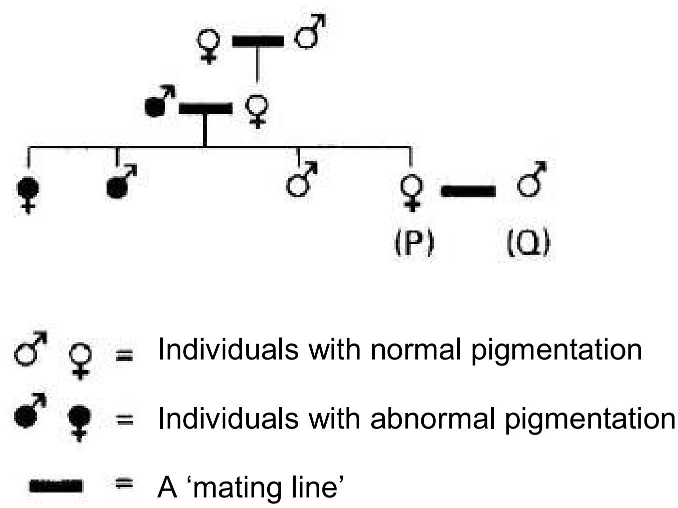
A Proto-oncogenes encode proteins which promote the cell cycle.

B Proto-oncogenes and tumor suppressor genes work the same way.

C When the expression of proto-oncogenes becomes aberrant, they can become oncogenes.

D At least one copy of a proto-oncogene needs to be mutated leading to a malignancy.

14 In a species where the female is homogametic, a sex-linked allelic pair of genes controls pigmentation. The following results were obtained during the course of a breeding experiment.



Which one of the following ratios of offspring will be produced when **P** and **Q** are bred together?

A 2 females (both normal): 2 males (both abnormal)

B 2 females (both carriers): 2 males (both normal)

C 2 females (1 carrier, 1 abnormal): 2 males (1 normal, 1 abnormal)

D 2 females (1 carrier, 1 normal): 2 males (1 normal, 1 abnormal)

15 Skin colour in onions is controlled by two pairs of alleles, **Ss** and **Rr**, which segregates independently. The allele **S** is dominant and must be present to allow development of pigment in the skin. In its absence, the onion skin is white. Allele **R** is dominant and gives a red colour; the recessive **r** gives a yellow colour.

What is the ratio of phenotypes in the offspring of a cross between plants of genotypes **SsRR** and **ssrr?**

A All red

B 1 red : 1 white

C 3 red : 1 white

D 1 white : 2 red : 1 yellow

16 A strain of toad has only one nucleolus in the nucleus of each cell instead of the usual two. When toads with one nucleolus per cell are mated, approximately a quarter of the offspring have two nucleoli per nucleus, half have one nucleolus per nucleus and a quarter have no nucleoli.

What is the most likely explanation of these results?

A The possession of one nucleolus is due to autosomal linkage.

B The possession of one nucleolus is due to the heterozygous condition.

C The allele for the presence of two nucleoli is recessive.

D The allele for the presence of two nucleoli is dominant.

17 In mice, two genes code for hair colour and length. These genes assort independently. The alleles are represented below:

Gene for hair colour: **B** black **b** brown	Gene for hair length: **L** short **l** long

A mouse with short, black hair was crossed with a mouse with long, brown hair. Half of the resulting offspring had short hair and half of the offspring had black hair.The genotype of the parent mouse with short, black hair must have been

A LLBB.

B LLBb.

C LlBB.

D LlBb.

18 Andrew, your classmate, has never had much energy. He goes to a doctor for help and is sent to the hospital for some tests. There they discover his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal.

Which of the following is the best explanation for his condition?

A His mitochondria lack the transport protein that moves pyruvate across the inner mitochondrial membrane.

B His cells cannot move NADH from glycolysis into the mitochondria.

C His cells contain an inhibitor that prevents oxygen use in his mitochondria.

D His cells have a defective electron transport chain, so glucose is reduced to lactate instead of to acetyl CoA.

19 Which statement on chemiosmotic mechanism of the coupling of substrate oxidation to the phosphorylation of ADP is **INCORRECT**?

A The main protein complexes of the electron transport chain in mitochondria pump protons out into the intermembranal space.

B A central role is played by a proton gradient that exists across the inner mitochondrial membrane.

C The pH inside the mitochondrial matrix is less than that in the cell cytoplasm.

D The electrochemical 'force' which drives the synthesis of ATP is called the proton-motive force.

20 The weedkiller DCMU blocks the flow of electrons chains in photophosphorylation. Why does this kill the plant?

A ATP and reduced NADP are not produced.

B ATP and reduced NAD are not produced.

C Photoactivation of the chlorophyll cannot occur.

D Photolysis of water does not occur.

21 Some photosynthetic organisms containing chloroplasts that lack PS II (photosystem II) are able to survive. The best way to detect the lack of PS II in these organisms would be to:

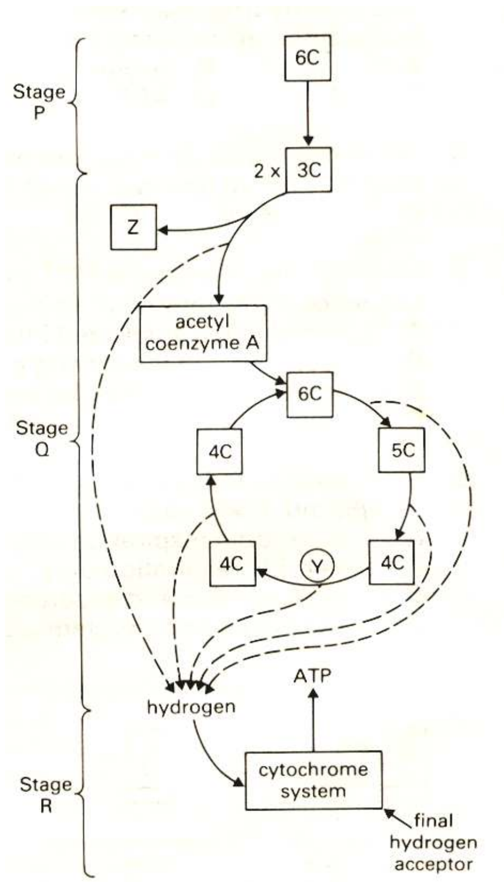
A determine if they have thylakoids in the chloroplasts.

B test for liberation of oxygen in the light.

C test for carbon fixation in the dark.

D determine the production of starch.

22 The diagram below shows the aerobic pathway in cellular respiration.



Which of the following indicates the correct locations at which stages P, Q and R occur in a cell?

	Cristae	Cytosol	Matrix
A	R	P	Q
B	Q	P	R
C	P	R	Q
D	R	Q	P

23 Which sequence of events correctly describes evolution?

I Differential reproduction of the spiders occurs.

II A new selection pressure occurs.

III Allele frequencies within the spider population change.

IV Poorly adapted spiders have decreased survivorship.

A II, IV, I, III

B II, IV, III, I

C IV, I, III, II

D IV, III, I, II

24 Modifications in the organisation of the basic pentadactyl limb (having 5 limbs) structure found invertebrates provides good evidence for the principle of

A adaptive radiation.

B genetic drift.

C convergent evolution.

D inheritance of acquired characters.

25 Which of the following statements could not be used to describe a species?

A A group of organisms showing distinctly similar autosomes.

B A group of organisms showing analogous body structures.

C A group of organisms capable of mating to produce viable offspring.

D A group of organisms sharing the same ecological niche.

26 Which of the following correctly explains why it is more advantageous to use yeast cells rather than bacterial cells for expression of eukaryotic DNA?

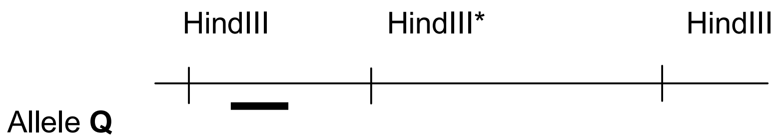
A Yeast cells can be cultured more readily than bacterial cells.

B Yeast cells can excise introns from the RNA transcript, but bacterial cells cannot.

C Yeast cells can reproduce more rapidly than bacterial cells.

D Yeast cells can produce proteins at a faster rate than bacterial cells.

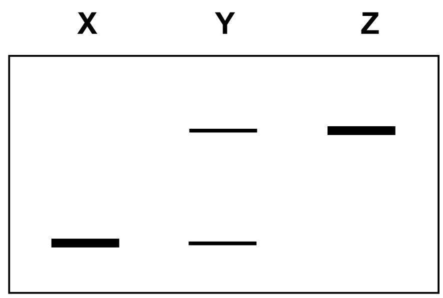
27 The region of the genome containing the RFLP used in this analysis is shown below.



HindIII indicates the restriction sites for this enzyme and * indicates the polymorphic site which is missing in the recessive allele **q**.

The black bar indicates the position of the probe used to detect the RFLP.

DNA fragments from three different individuals, **X**, **Y** and **Z**, were subjected to restriction digestion by HindIII and separate by gel electrophoresis. The following results were obtained.



Deduce the genotype of individual **Z**.

A **QQ**

B **Qq**

C **qq**

D $\mathbf{X^{Q}Y}$

28 Which of the following is the most powerful way of increasing the specificity of a DNA profile analysis?

A Select markers present on the sex chromosomes rather than on the autosomes.

B Analyze each marker by PCR rather than RFLP analysis.

C Repeat the analysis multiple times.

D Increase the number of markers used.

29 Which of the following statements is false about hematopoietic stem cells?

A Hematopoietic stem cells can differentiate into all specialised cells.

B Hematopoietic stem cells can differentiate into all blood cell types.

C Hematopoietic stem cells can be found in the bone marrow.

D Hematopoietic stem cells are able to reproduce continually.

30 Plants are more readily manipulated by genetic engineering than are animal cells because

A plant genes do not contain introns.

B more vectors are available for transferring recombinant DNA into plant cells.

C a somatic plant cell can often give rise to a complete plant.

D plant cells have larger nuclei.

End of H1 Paper 1